Conception phase

1. Abstract

This database design conceptualizes a scalable accommodation booking platform based on the Airbnb use case. The data model addresses the need for a normalized, robust structure to handle complex interactions between hosts, guests, and listings.

The solution creates a schema of **23 entities** to ensure data integrity and minimize redundancy. Key design features include:

1. **Advanced Categorization:** Properties are classified using both marketing categories (via product_category) and functional attributes (via property_attribute), allowing for precise filtering.
2. **Financial Accuracy:** All monetary values use DECIMAL(10,2) types to support exact commission and fee calculations.
3. **Normalized Hierarchies:** Location data is separated into Location, Country, and Region tables to support global scalability.

This conception phase establishes the theoretical foundation for the SQL implementation, defining all necessary primary/foreign key relationships and many-to-many join tables required for a fully functional booking system.

2. Requirements Specification

2.1 Roles (User Groups)

- **User/Guest:** Can search for properties, filter by region, categories, or amenities, and make bookings. They can "favorite" properties and leave reviews.
- **Host:** A user who lists properties. They have a specific "host start date" and can list multiple languages they speak.
- **System/Admin:** Manages the standardized "lookup" lists (e.g., property types, guest types, review components) to ensure data consistency.

2.2 Actions & Functionalities

- **Search & Discovery:** Users can filter properties by **Location** (Region/Country) or by **Category** (e.g., "Amazing view", "Countryside").
- **Detailed Property Listing:** Properties display specific **Attributes** (amenities like Wifi) categorized by type (e.g., "Safety", "House Rules").
- **Booking Workflow:** Users select dates and guest counts (adults/children/pets). The system calculates the total price including fees and tracks the booking status (e.g., "Approved").
- **Review System:** Reviews include an overall rating and granular ratings for specific components (e.g., Cleanliness), managed via a joining table.

2.3 Data Requirements

- **Financial Precision:** Prices and fees are stored using DECIMAL(10,2) to handle exact currency values.
- **Normalization:** Address data is split into address_line fields while broader location data is normalized into Location, Country, and Region entities.
- **Media:** Property images are stored with URLs and a specific display order (img_order).

3. Entity Relationship Model (ERM)

3.1 List of Entities

The data model consists of the following 23 entities, fully covering the system requirements:

1. **property**: Core entity storing price, description, and address details.
2. **location**: Lookup for cities/locations.
3. **country**: Lookup for countries.
4. **region**: Lookup for global regions (e.g., Europe).
5. **image**: Stores photo URLs and display order.

6. **user_account**: Stores user details (Name, Email, Password).
7. **favourite**: Joining table for Users <=> Properties.
8. **category**: Marketing categories like "OMG" or "Amazing Pools".
9. **product_category**: Joining table linking Properties to Categories.
10. **attribute**: Lookup for amenities (e.g., "Wifi", "Kitchen").
11. **attribute_category**: Categories for attributes (e.g., "Safety", "House Rules").
12. **property_attribute**: Joining table linking Properties to Attributes.
13. **property_type**: Lookup for "House", "Apartment", "Hotel".
14. **place_type**: Lookup for "Entire Place" vs. "Private Room".
15. **booking**: Stores dates, fees, and total price.
16. **booking_status**: Lookup for statuses like "Approved", "Cancelled".
17. **guest_type**: Lookup for "Adult", "Child", "Infant".
18. **guest_booking**: Joining table counting guests by type per booking.
19. **user_review**: Main review record with text and date.
20. **review_component**: Lookup for rating aspects like "Cleanliness".
21. **component_rating**: Joining table for specific component scores.
22. **language**: Lookup for languages spoken.
23. **user_language**: Joining table for Users <=> Languages.

3.2 Triple Relationships (Examples)

- **Location Hierarchy:** property → location → country → region.
- **Booking Guests:** booking → guest_booking → guest_type.
- **Property Attributes:** property → property_attribute → attribute (linked to attribute_category).

4. Data Dictionary

4.1 Location & Geography

TABLE	ATTRIBUTE	DATA TYPE	DESCRIPTION
REGION	id	INT (PK)	Unique identifier for the global region.
	region_name	VARCHAR(255)	Name of the region (e.g., "Europe"). This field must be NOT NULL and UNIQUE
COUNTRY	id	INT (PK)	Unique identifier for the country.
	region_id	INT (FK)	Links the country to a specific global region.
	country_name	VARCHAR(255)	Name of the country (e.g., "Germany"). This field must be NOT NULL and UNIQUE
LOCATION	id	INT (PK)	Unique identifier for the location/city.
	country_id	INT (FK)	Links the location to a specific country.

	location_name	VARCHAR(255)	Name of the city or area (e.g., "Berlin").

4.2 Property Details

TABLE	ATTRIBUTE	DATA TYPE	DESCRIPTION
PROPERTY	id	INT (PK)	Unique identifier for the property listing.
	host_id	INT (FK)	The user ID of the property owner/host.
	location_id	INT (FK)	The location ID where the property is situated.
	place_type_id	INT (FK)	Type of place (e.g., Entire place vs. Room).
	property_type_id	INT (FK)	Type of structure (e.g., Apartment vs. House).
	night_price	DECIMAL(10,2)	Cost per night in local currency.
	property_name	VARCHAR(100)	Title of the listing.
	num_guests	TINYINT	Max guests allowed.
	num_bed	TINYINT	Number of beds available.
	num_bathroom	TINYINT	Number of bathrooms.
	is_guest_favourite	TINYINT	Flag indicating if it is a top-rated listing.
	description	TEXT	Full description of the property.
	address_line1	VARCHAR(255)	Detailed street address.
	address_line2	VARCHAR(255)	Detailed street address.
PROPERTY_TYPE	id	INT (PK)	Unique ID for property structure type.
	type_name	VARCHAR(50)	Name (e.g., "Apartment", "House", "Hotel").This field must be NOT NULL and UNIQUE
PLACE_TYPE	id	INT (PK)	Unique ID for privacy type.
	type_name	VARCHAR(100)	Name (e.g., "Entire place", "Private room").This field must be NOT NULL and UNIQUE
IMAGE	id	INT (PK)	Unique identifier for the image.
	property_id	INT (FK)	The property this image belongs to.
	img_url	VARCHAR(2048)	URL path to the image file.
	img_order	TINYINT	Display order in the gallery (e.g., 1 = Cover).

4.3 User Management

TABLE	ATTRIBUTE	DATA TYPE	DESCRIPTION
USER_ACCOUNT	id	INT (PK)	Unique user identifier.
	first_name	VARCHAR(50)	User's first name.
	last_name	VARCHAR(50)	User's last name.
	email	VARCHAR(255)	User's login email address. This field must be NOT NULL and UNIQUE
	password	VARCHAR(255)	User's hashed password.
	join_date	DATETIME	Timestamp when the user registered.
	date_host_started	DATE	Date the user became a host (can be null).

LANGUAGE	**ID**	**INT (PK)**	**UNIQUE LANGUAGE ID.**
	language_name	VARCHAR(50)	Name of language (e.g., "English", "German"). This field must be NOT NULL and UNIQUE
USER_LANGUAGE	user_id	INT (FK)	Link to User table.
	language_id	INT (FK)	Link to Language table.
FAVOURITE	user_id	INT (FK)	Link to the User who liked the property.
	property_id	INT (FK)	Link to the Property being liked.

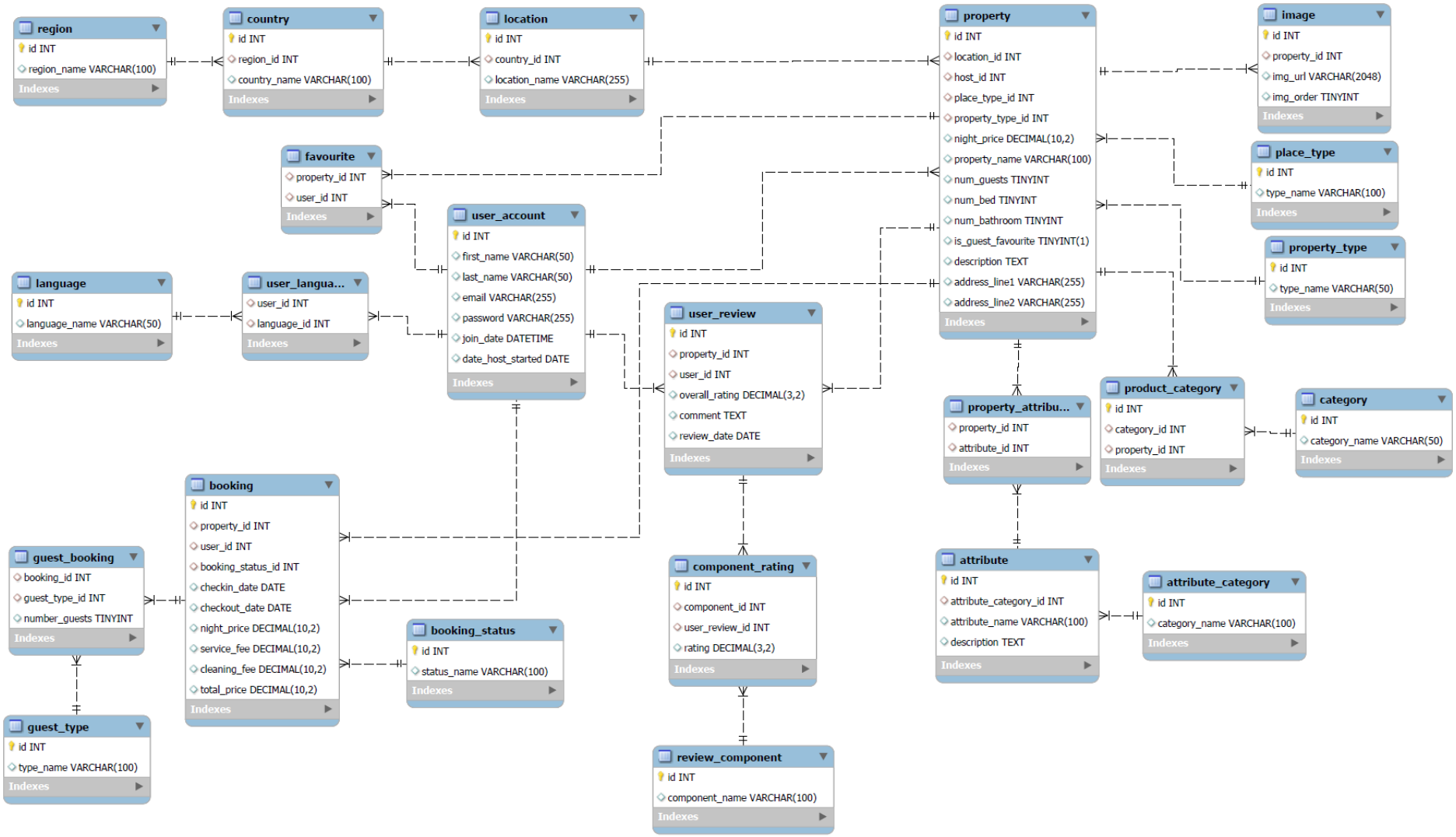
4.4 Attributes & Categories

TABLE	ATTRIBUTE	DATA TYPE	DESCRIPTION
CATEGORY	id	INT (PK)	Unique marketing category ID.
	category_name	VARCHAR(50)	Name (e.g., "OMG", "Countryside"). This field must be NOT NULL and UNIQUE
PRODUCT_CATEGORY	id	INT (PK)	Unique ID for this relationship.
	category_id	INT (FK)	The category being assigned.
	property_id	INT (FK)	The property receiving the category.
ATTRIBUTE_CATEGORY	id	INT (PK)	Unique ID for attribute groups.
	category_name	VARCHAR(100)	Group name (e.g., "Amenities", "Safety"). This field must be NOT NULL and UNIQUE
ATTRIBUTE	id	INT (PK)	Unique attribute ID.
	attribute_category_id	INT (FK)	Group this attribute belongs to.
	attribute_name	VARCHAR(100)	Name (e.g., "Wifi", "Kitchen").
	description	TEXT	Optional description of the amenity.
PROPERTY_ATTRIBUTE	property_id	INT (FK)	The property having the amenity.
	attribute_id	INT (FK)	The amenity being assigned.

4.5 Bookings

TABLE	ATTRIBUTE	DATA TYPE	DESCRIPTION
BOOKING	id	INT (PK)	Unique reservation ID.
	property_id	INT (FK)	The property being booked.
	user_id	INT (FK)	The guest making the booking.
	booking_status_id	INT (FK)	Current status of the reservation.
	checkin_date	DATE	Start dates of the stay.
	checkout_date	DATE	End dates of the stay.
	night_price	DECIMAL(10,2)	Snapshot of nightly rate at time of booking.
	service_fee	DECIMAL(10,2)	Associated fees.
	cleaning_fee	DECIMAL(10,2)	Associated fees.
	total_price	DECIMAL(10,2)	Final calculated price.
BOOKING_STATUS	id	INT (PK)	Unique ID for status.
	status_name	VARCHAR(100)	Status (e.g., "Confirmed", "Cancelled").This field must be NOT NULL and UNIQUE
GUEST_TYPE	id	INT (PK)	Unique ID for guest type.
	type_name	VARCHAR(100)	Type (e.g., "Adult", "Child", "Infant").This field must be NOT NULL and UNIQUE
GUEST_BOOKING	booking_id	INT (FK)	The booking record.
	guest_type_id	INT (FK)	The type of guest.
	number_guests	TINYINT	Count of guests for this type.

4.6 Reviews

TABLE	ATTRIBUTE	DATA TYPE	DESCRIPTION
USER_REVIEW	id	INT (PK)	Unique review ID.
	property_id	INT (FK)	The property being reviewed.
	user_id	INT (FK)	The guest writing the review.
	overall_rating	DECIMAL(3,2)	Aggregate score (e.g., 4.85).
	comment	TEXT	Written feedback.
	review_date	DATE	Date the review was posted.
REVIEW_COMPONENT	id	INT (PK)	Unique ID for rating aspect.
	component_name	VARCHAR(100)	Aspect name (e.g., "Cleanliness", "Accuracy").This field must be NOT NULL and UNIQUE
COMPONENT_RATING	id	INT (PK)	Unique ID for this specific rating.
	user_review_id	INT (FK)	The parent review.
	component_id	INT (FK)	The aspect being rated.
	rating	DECIMAL(3,2)	Score given for this aspect.

5 Figure 1: Complete Entity Relationship Model of the Booking System